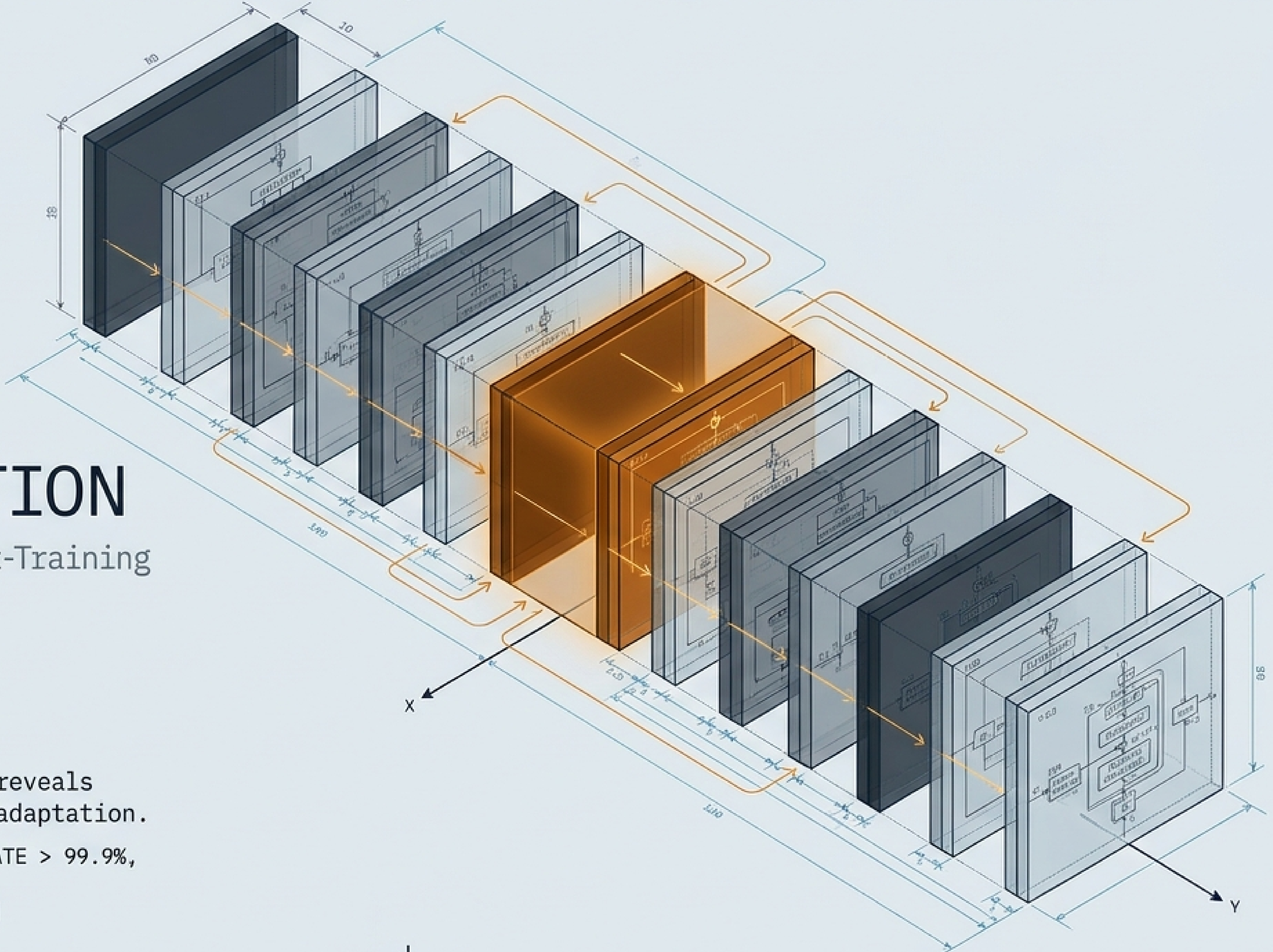
REPAIR, NOT RECONSTRUCTION

The Architecture of RL Post-Training



What the One Layer phenomenon reveals about the structure of neural adaptation.

[ADAPTATION_METRICS: ADAPTATION_RATE > 99.9%,
STRUCTURAL_INTEGRITY: PRESERVED,
FOCAL_POINT: LAYER_6_ACTIVATION]

THE EMPIRICAL ANOMALY: IS ONE LAYER ENOUGH?

THE SCALE

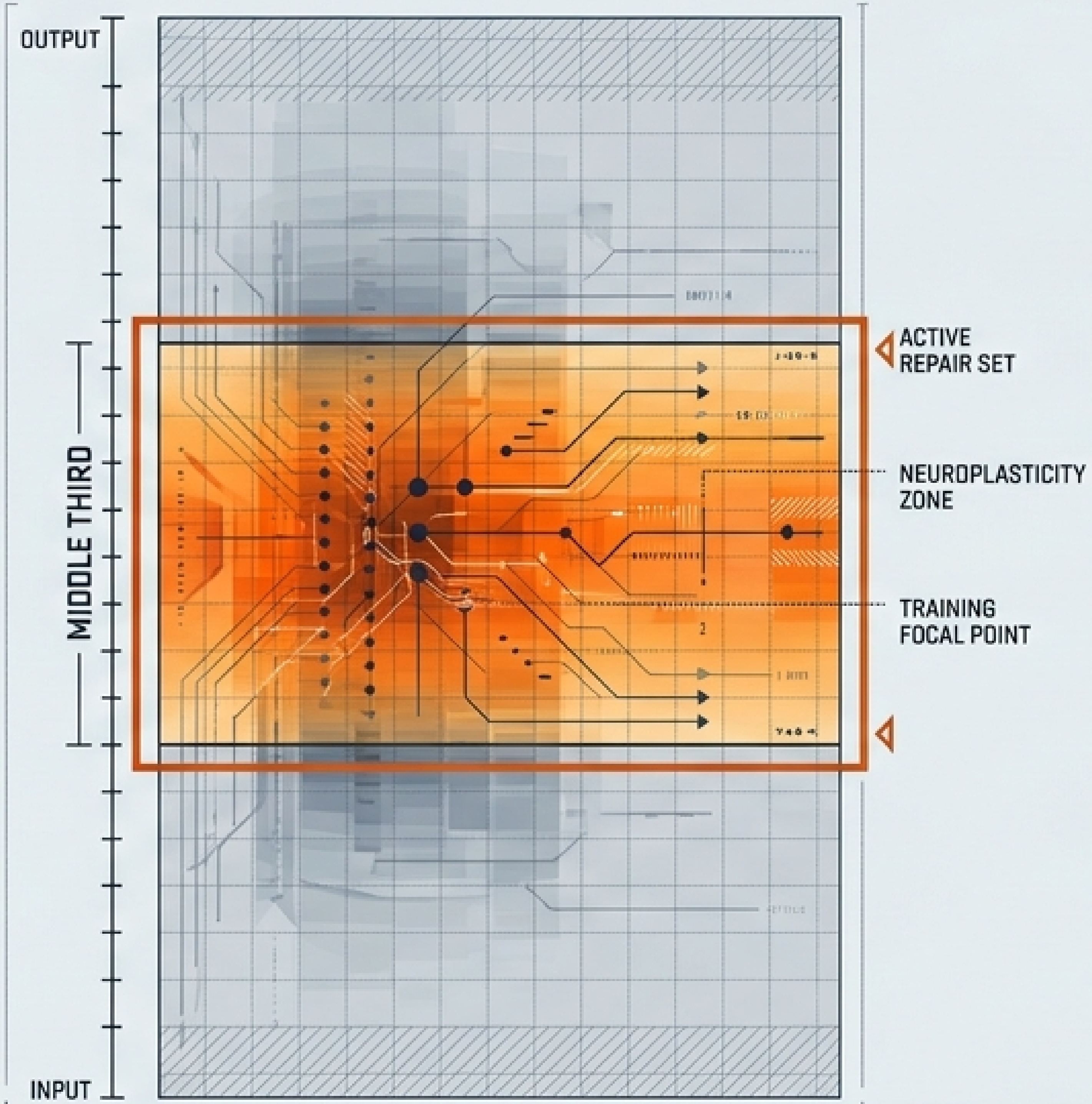
7 MODELS

2 FAMILIES

3 RL ALGORITHMS

3 TASK DOMAINS

THE FINDING

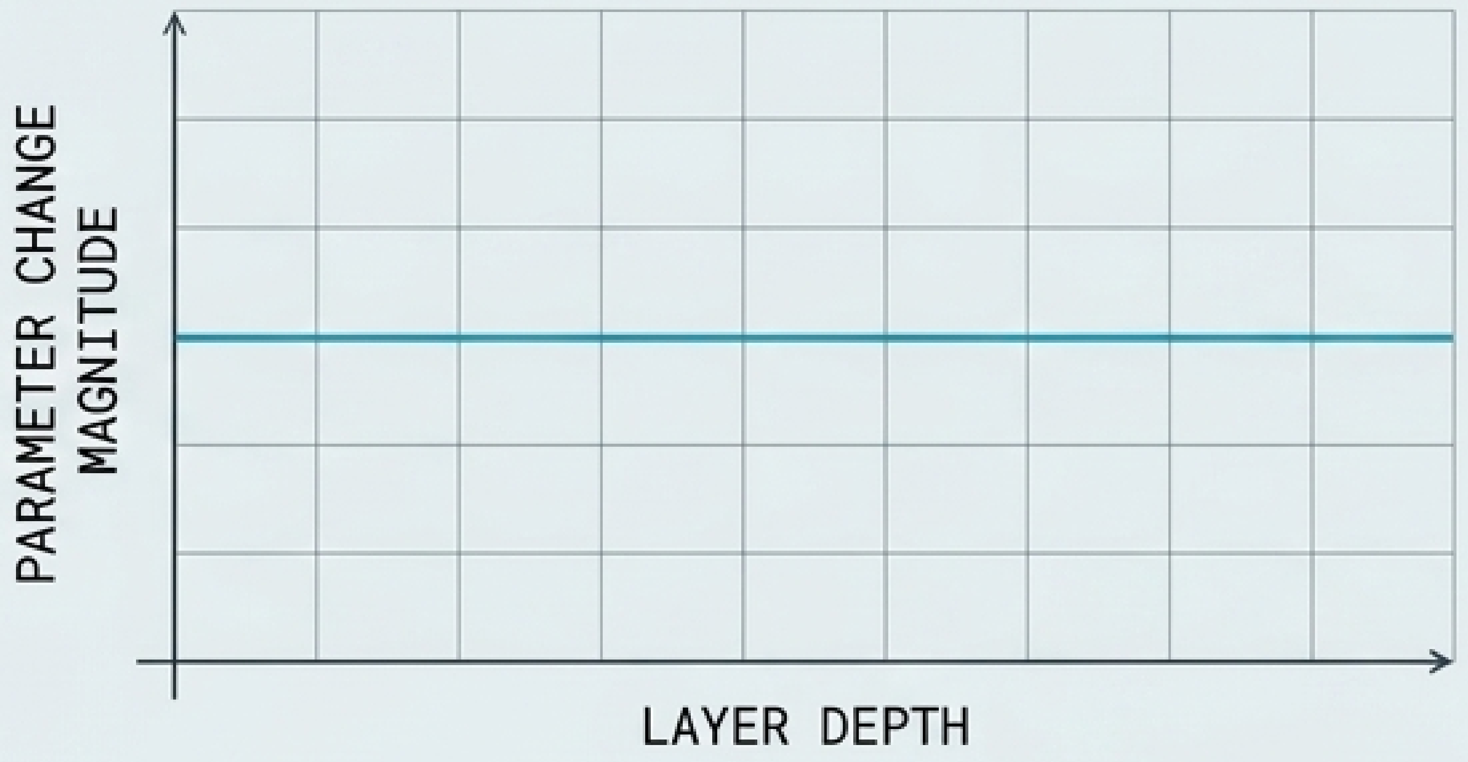


FREEZING THE ENTIRE MODEL AND TRAINING ONLY **ONE LAYER** IN THE MIDDLE UNDER RL RECOVERS—AND SOMETIMES EXCEEDS—FULL-PARAMETER TRAINING PERFORMANCE.

SOURCE: ZHANG ET AL., 2026

RULING OUT THE OBVIOUS EXPLANATIONS

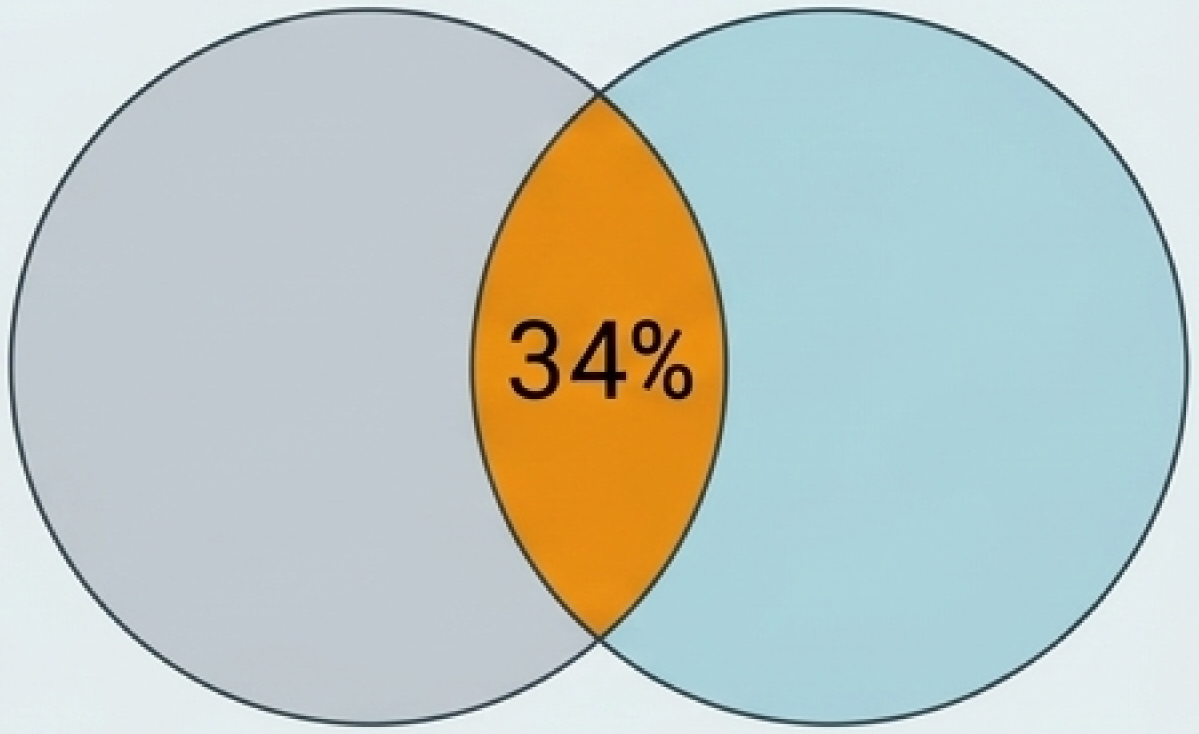
WEIGHT DISPLACEMENT



Hypothesis: Middle layers move more.

Reality: Parameter displacement is perfectly uniform across depth.

SOLUTION OVERLAP



Hypothesis: Middle layers learn the exact same thing.

Reality: The top 7 highest-contribution layers show only 34% pairwise overlap in newly solved problems.

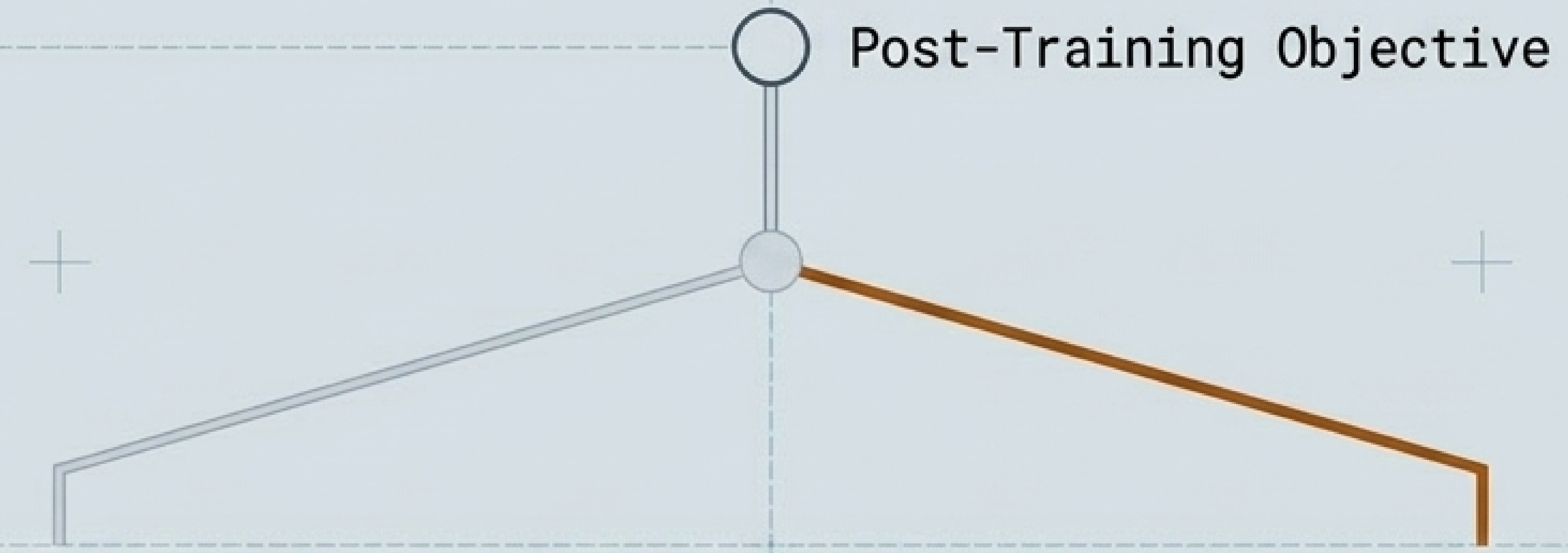
THE MIDDLE-LAYER CONCENTRATION IS A GEOMETRIC PROPERTY OF THE PARAMETER SPACE, NOT A BYPRODUCT OF UPDATE SIZE OR REDUNDANT LEARNING.

THE FLAW OF UNIFORM TRAINABILITY

	REPRESENTATIONAL NECESSITY (FORWARD PASS)	ADAPTIVE USEFULNESS (RL TRAINING)
DEFINITION	Which parameters the system currently depends on for behavior.	Which parameters can absorb a training signal to improve behavior.
MEASUREMENT	Ablation and Removal studies.	Selective parameter freezing and training.
OUTCOME	Widespread across the network depth.	**Highly concentrated in the middle third.**

Prior work assumes adaptation is distributed in rough proportion to representational importance. RL post-training proves this false: the parameters a system depends on are not necessarily the ones that offer useful directions for adaptation.

RE-DEFINING REPAIR IN COMPLEX SYSTEMS



RECONSTRUCTION

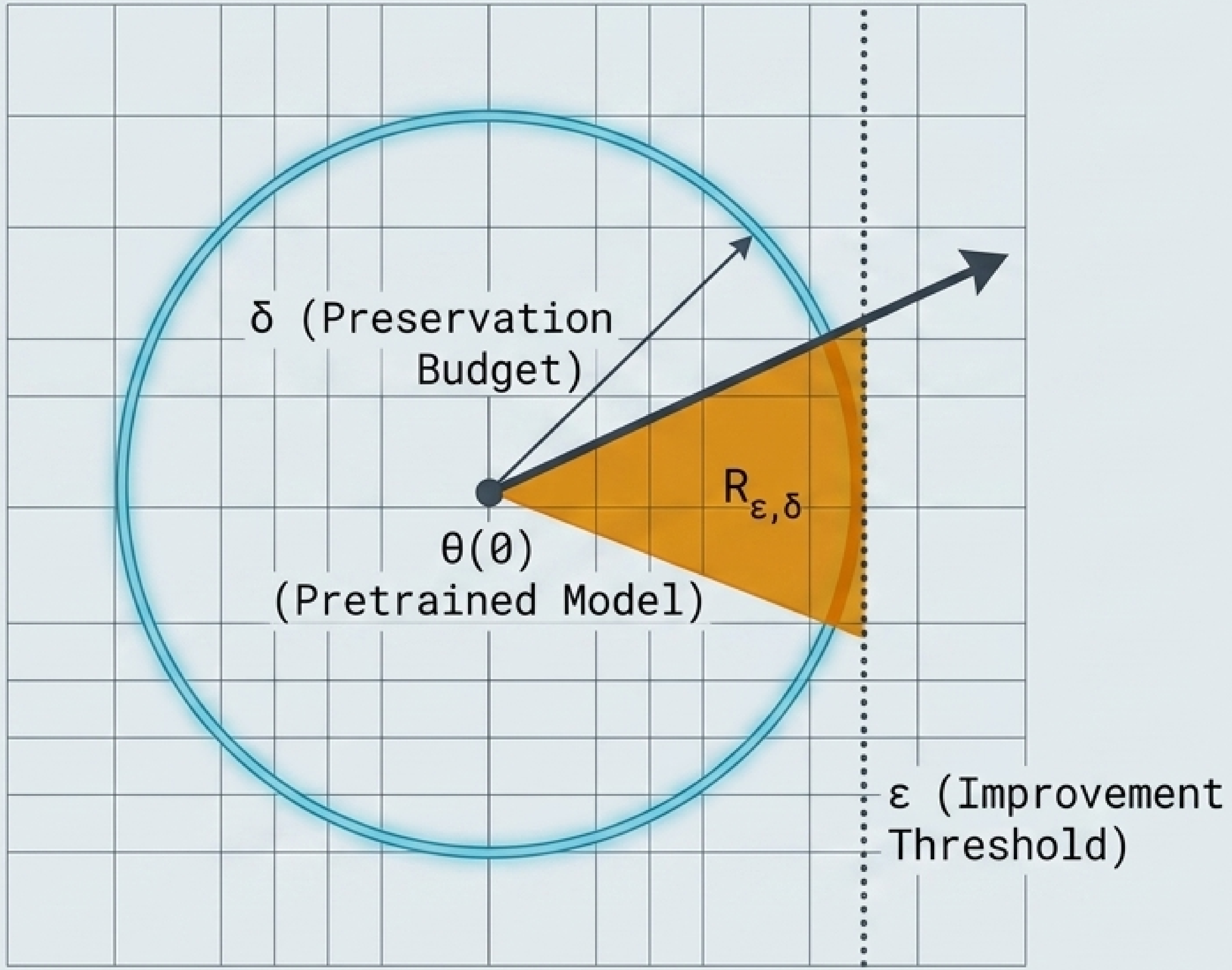
Restoring a defective system to a prior standard from the ground up.

REPAIR

Adding a new capability without unraveling the delicate, inherited organization of a functioning system.

RL post-training does not begin from an undifferentiated parameter space. It begins from a pretrained point that already embodies a coherent organization. Optimization is heavily constrained by this pre-existing structure.

THE LOCAL REPAIR SET FORMALISM



δ = Preservation Budget

Maximum allowable damage to current inherited capabilities.

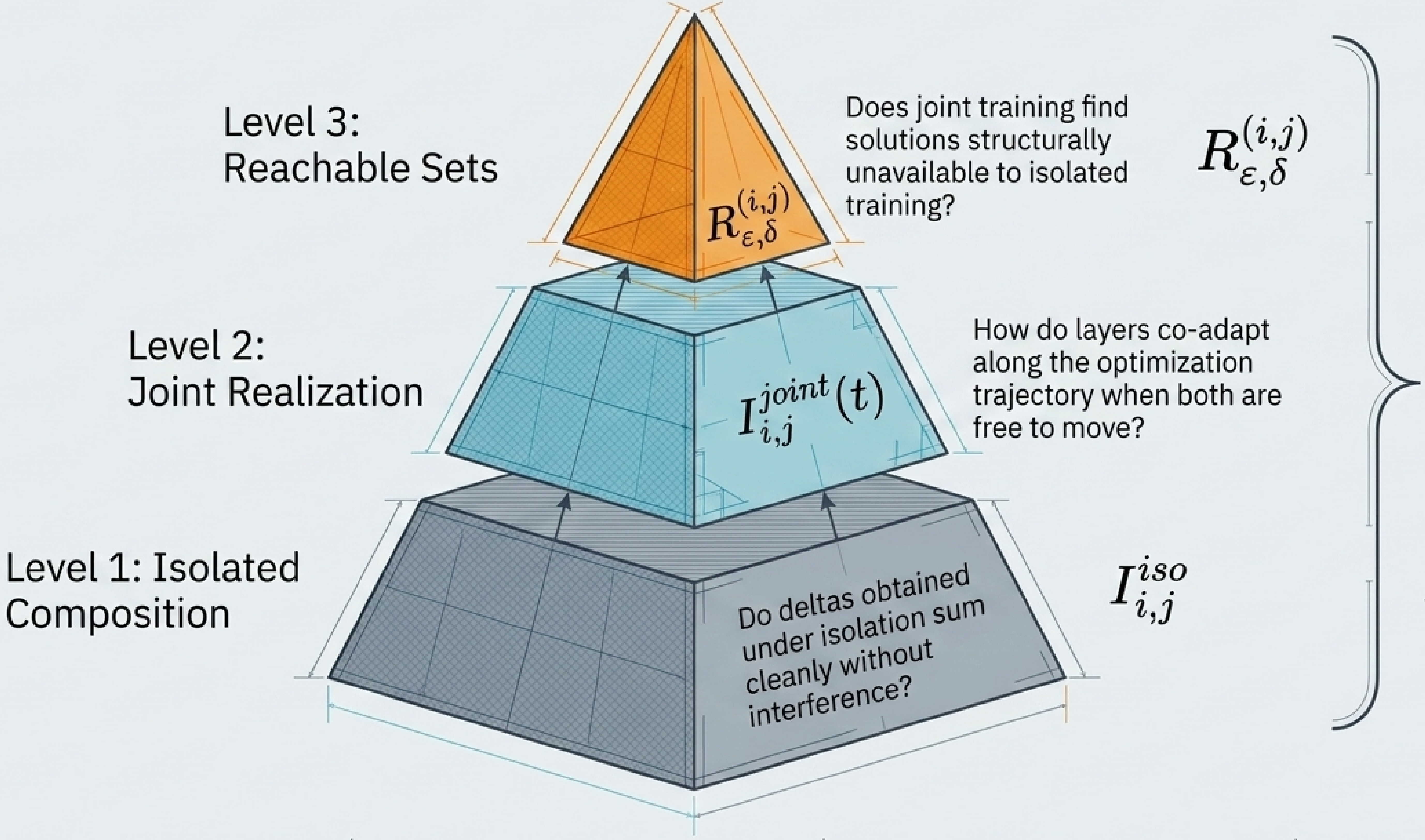
ε = Improvement Threshold

Minimum required task gain in the RL objective.

$R_{\varepsilon,\delta}$ = The Repair Set

An update must satisfy the joint criterion: moving a layer must lower the loss while strictly respecting the bounds of the preservation budget.

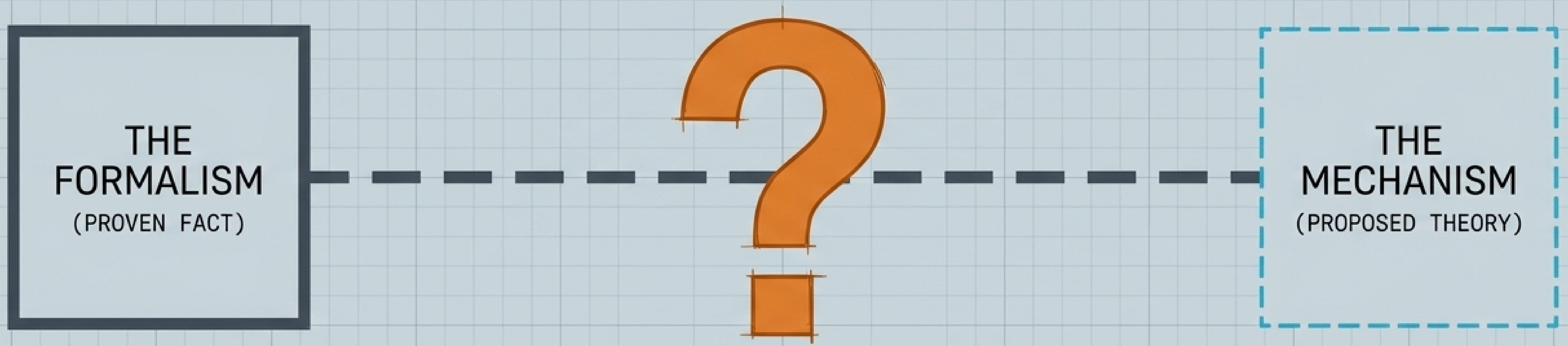
The Interaction Hierarchy: Do Repairs Compose?



These levels are not substitutes. A failure at Tier 1 does not mean co-adaptation at Tier 2 will fail.

Full-parameter training explicitly searches the Apex.

THE EXPLANATORY GAP

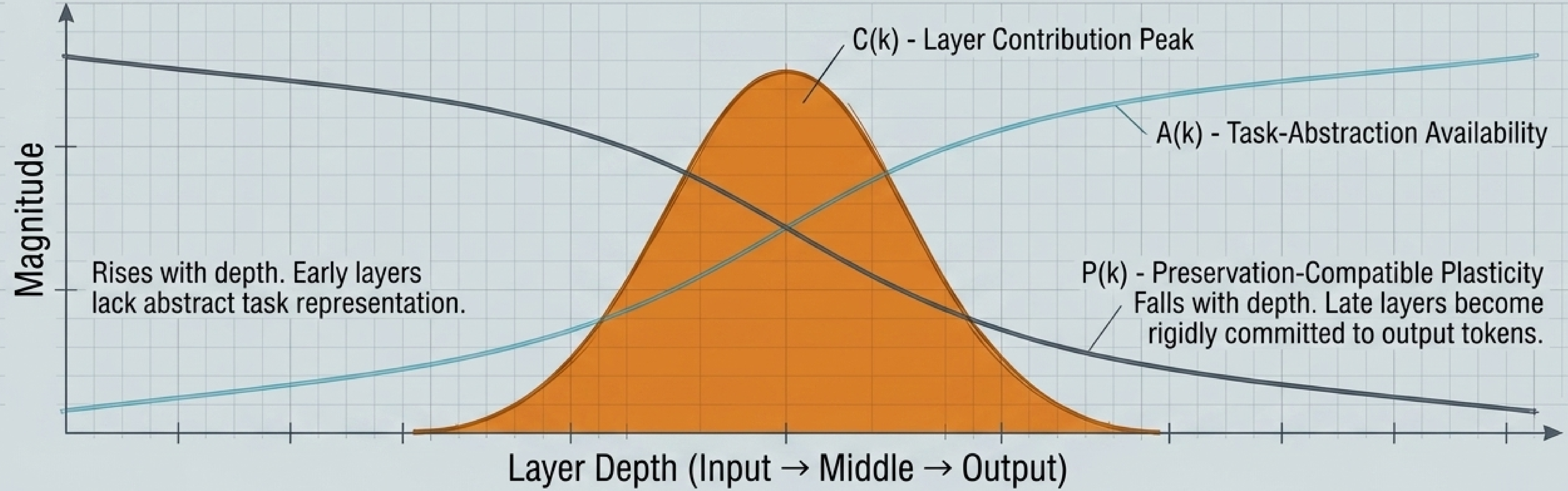


THE FORMALISM TELLS US HOW TO MEASURE THE REPAIR SET. BUT WHY DO THE USABLE, HIGHLY-DENSE REPAIR SETS CONCENTRATE STRICTLY IN THE MIDDLE OF THE NETWORK?

Transitioning from the measurable findings to a speculative mechanism grounded in iterative refinement and transformer interpretability.

The Product Hypothesis

$$C(k) \propto A(k) \times P(k)$$



Usable repair directions peak at intermediate depth: early enough that the residual stream can still absorb a perturbation, but late enough that the representation actually encodes the task.

Operationalizing the Preservation Budget

$$D_{pres} = \lambda_{beh} D_{beh} + \lambda_{rep} D_{rep} + \lambda_{int} D_{interface}$$

Behavior (D_{beh})

Degradation on inherited capabilities the modification was not intended to affect.

Measured via hold-out benchmarks independent of RL training.

Representation (D_{rep})

Divergence of internal hidden-state representations from the pretrained model.

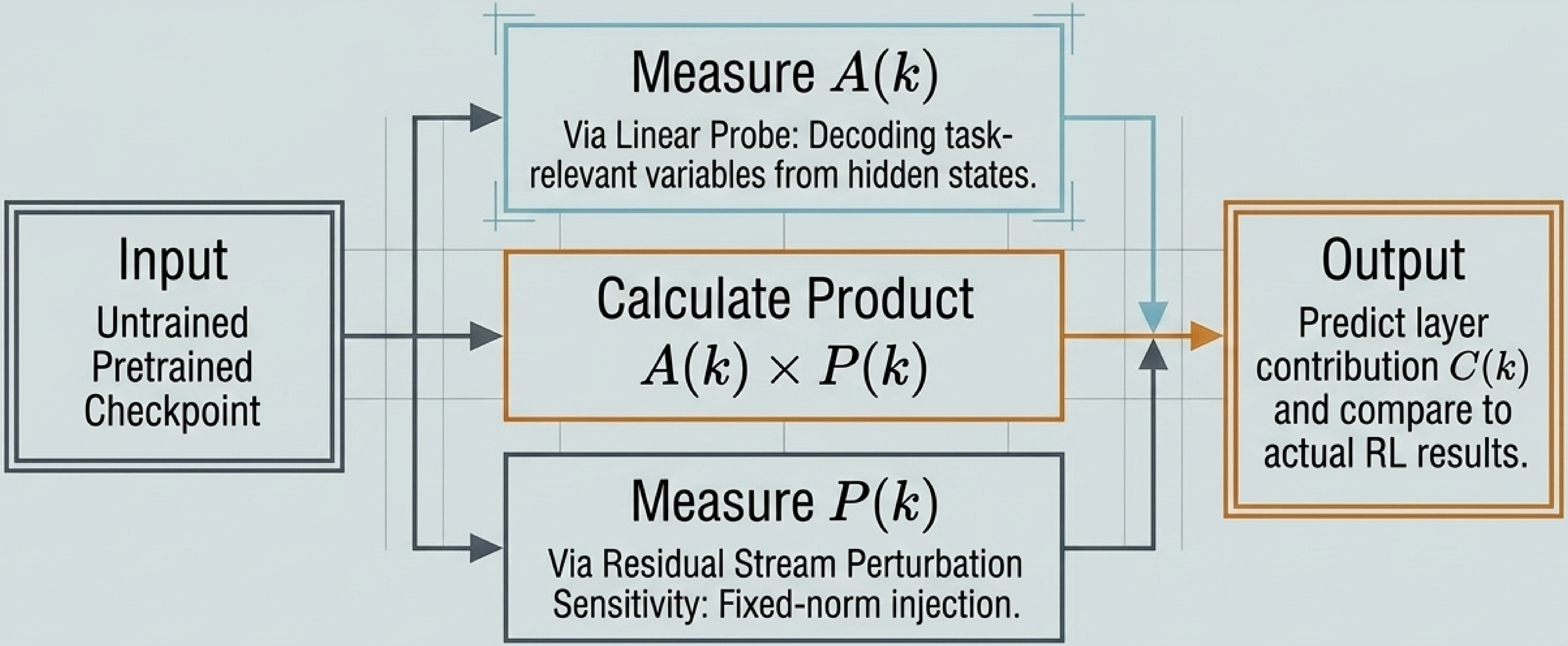
Interface ($D_{interface}$)

Disturbance specifically to the input-encoding and output-decoding conventions.

The architectural standard the rest of the network depends upon.

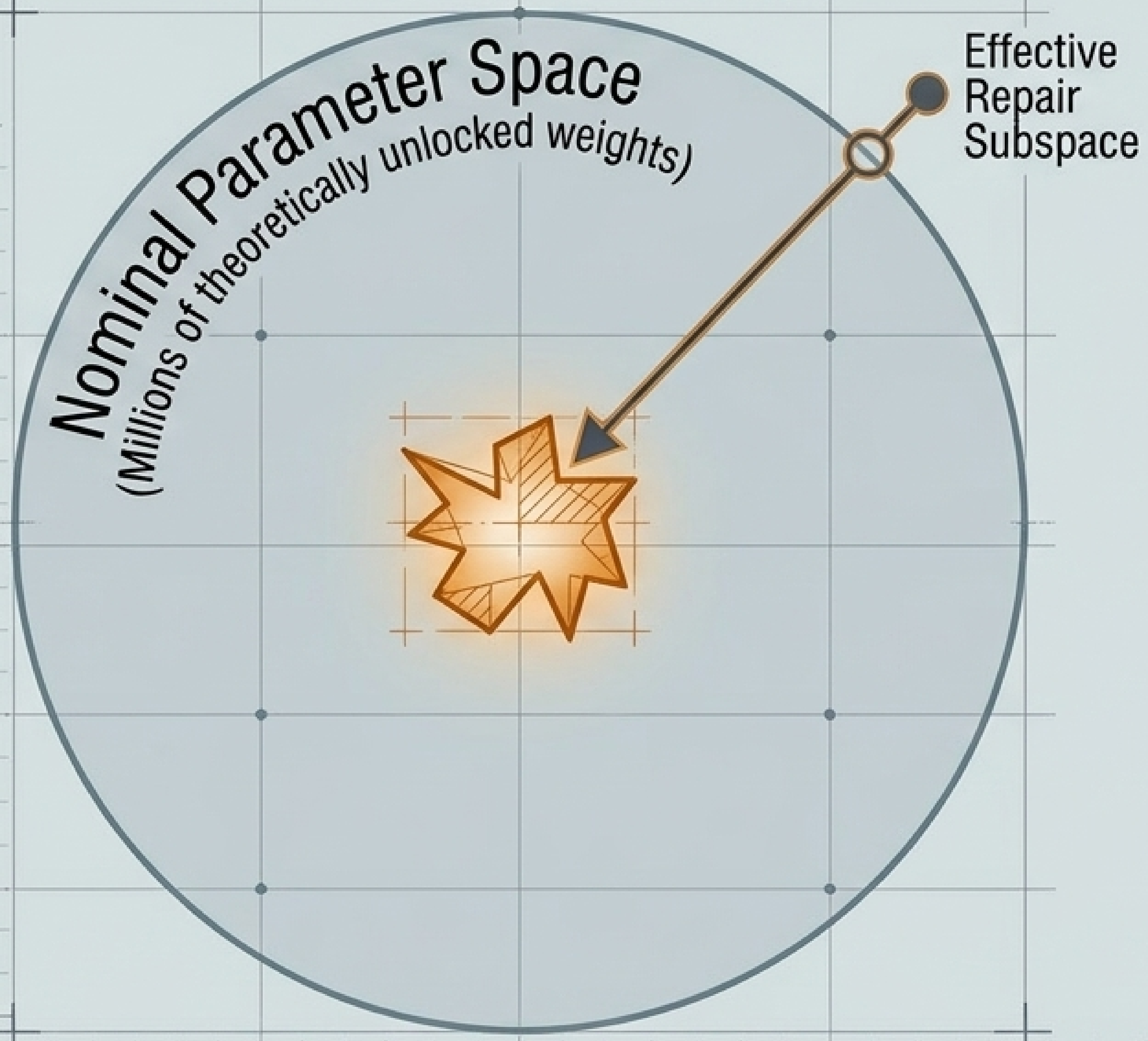
A valid repair must minimize damage across these three distinct structural axes without unraveling pre-existing organization.

The Falsifiable Base-Model Test



The beauty of the hypothesis: its components can be extracted entirely from the geometry of the pretrained model before any reinforcement learning takes place.

Nominal vs. Effective Freedom



A **system's nominal freedom** to be **modified** is **not the same** as its **effective freedom** to be improved.

Optimization is not a free exploration of mathematical space. It is tightly bound inside a pre-existing organization. Most nominally available modifications are disruption dressed as improvement.

The Roadmap Forward

1. The Base-Model Test	2. Static Composability	3. Joint Trajectory Logging
Run the $A(k) \times P(k)$ linear probe and perturbation tests on public checkpoints to validate the Product Hypothesis without RL training.	**Calculate** $I_{i,j}^{\text{iso}}$ directly from existing single-layer checkpoint deltas to map **independent repair compatibility** across all layers.	Run targeted multi-layer joint training to measure $I_{i,j}^{\text{joint}}(t)$ and determine if **co-adaptation unlocks new reachable sets**.

These experiments are designed to fail independently. Validating the mechanism leaves the formal distinction between nominal and effective modification spaces intact.